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3 OT Zero Trust Defenses

for SMB Manufacturing

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Do hackers see SMB manufacturing as **"low-hanging fruit"?**

71% **of small businesses are already targets of ransomware.**[1]

As a small to medium business owner, perhaps you found a niche market and became a specialist project shop that makes custom equipment. You made an investment in some manufacturing equipment that uses a version of Windows that is no longer supported, so you don't receive any security patches or updates. But the equipment still works and it would be costly to replace it.

Suppose you won the big order and developed a novel way to batch jobs into faster operations, allowing production of higher volumes. Now several large clients depend on you. You depend on your supply chain that includes a specialized device that is a key component of your product. How do you make sure this equipment is not infected with malware that you may inadvertently pass on to your customers?

Suppose you are rapidly becoming the producer of high volumes of in-demand products such as soda cans or auto parts. Maintaining the supply chain and logistics becomes even more important to keep your manufacturing processes moving.

IT software tools and anti-malware are available for small to medium size businesses. There are SaaS providers with IT cybersecurity built-in. So, you buy cybersecurity insurance. You enforce secure passwords and train your team not to click on phishing emails.

What are you missing? The fact is, your manufacturing equipment is still vulnerable because IT anti-malware was never designed to defend operational technology. That legacy equipment that can't be patched and those devices you embed in your product are vulnerable. Without specialized cybersecurity for your operational technology, your manufacturing processes are easy targets for bad actors looking to make a fast buck by extorting a ransom.

[1] *"How ransomware is a big problem for small business – and what to do about it," Insureon.com, July 2 2020, accessed June 11 2022.*

Harden endpoints before attackers can strike.

1

47% **of attacks on the manufacturing industry have exploited vulnerabilities that businesses either could not patch or failed to patch.[2]**

Many SMB manufacturers invest in equipment that could have a long, productive life. The catch is, that equipment's software vendors will have sometimes stopped sending security patches, leaving these machines open to attack. Now, you are faced with the dilemma of either spending your hard-earned profits to replace these systems or simply hoping that hackers don't find them.

Virtual patching stops attackers from exploiting the frequently unpatched or unpatchable status of legacy assets. Productivity is not disrupted by these defenses, either, which require no changes to assets being secured due to virtual patching being deployed at the network level. This functionality is included with next generation firewalls that are based on the OT zero trust methodology.

The industry verticals that are targeted most by ransomware attacks are construction, manufacturing and professional services[3]. The most common ransomware families currently attacking SMB manufacturing are CryptoLocker, CryptoWall, Locky and WannaCry. OT zero trust can protect you from all of these threats and more.

Ransomware attacks equipment referred to as "endpoints" – assets. OT zero trust endpoint protections adapt to any security need. The OT zero trust philosophy is never trust, always verify. Using deep trust analysis of the special protocols and machine types used in manufacturing systems, OT zero trust knows what's going on at any given moment. Trust lists are continually updated to allow only trustworthy control commands authorized by your employees to reach your equipment. When OT zero trust protectors discover an infected payload, they will promptly quarantine it and send you an alert. You can also see the current state of all your manufacturing systems on a single pane of glass, the OT defense console.

Zero day attacks occur when a new kind of malware turns up that equipment manufacturers have not had time to patch. OT zero trust zaps these unpredictable attacks by using virtual patching to put a network-based "shield" around vulnerable

[2] Graham Cluley, "Manufacturing was the top industry targeted by ransomware last year", Tripwire.com, Feb. 24 2022, accessed June 11 2022.
[3] "Ransomware is the Biggest Threat for Small to Medium Businesses," Security Magazine, June 3 2019, accessed June 11 2022.

assets. Threat intelligence is always being evaluated by OT zero trust machine learning systems to learn the popular attack vectors and provide the contingency controls to stop hacks before they happen.

Use a handheld rapid-scan device to cover inbound and air-gapped devices.

2

With OT zero trust-based handheld scan devices, your specialists can conduct fast scans of each asset coming onto your work site. This guarantees that malware doesn't make it onto your work site hidden in onboarding assets or in devices brought on-site by employees.

OT zero trust-based handheld scanners look like USB devices. Plug them in and they automatically perform the pre-defined setting – either wipe away malware automatically, or report its existence and stand by for further input.

When a device like this that is easily moved from place to place creates an inventory of each device scanned, that inventory can then be sent to a centralized management console for use as a dated proof of digital hygiene ("OT health check") or for further investigation. This supports outcomes such as identifying the precise tools necessary to stop malware, even when your equipment is running older versions of Windows or Linux, or much more easily discovering what weak links are getting cyber threats onto your shop floor.

Segment the network.

3

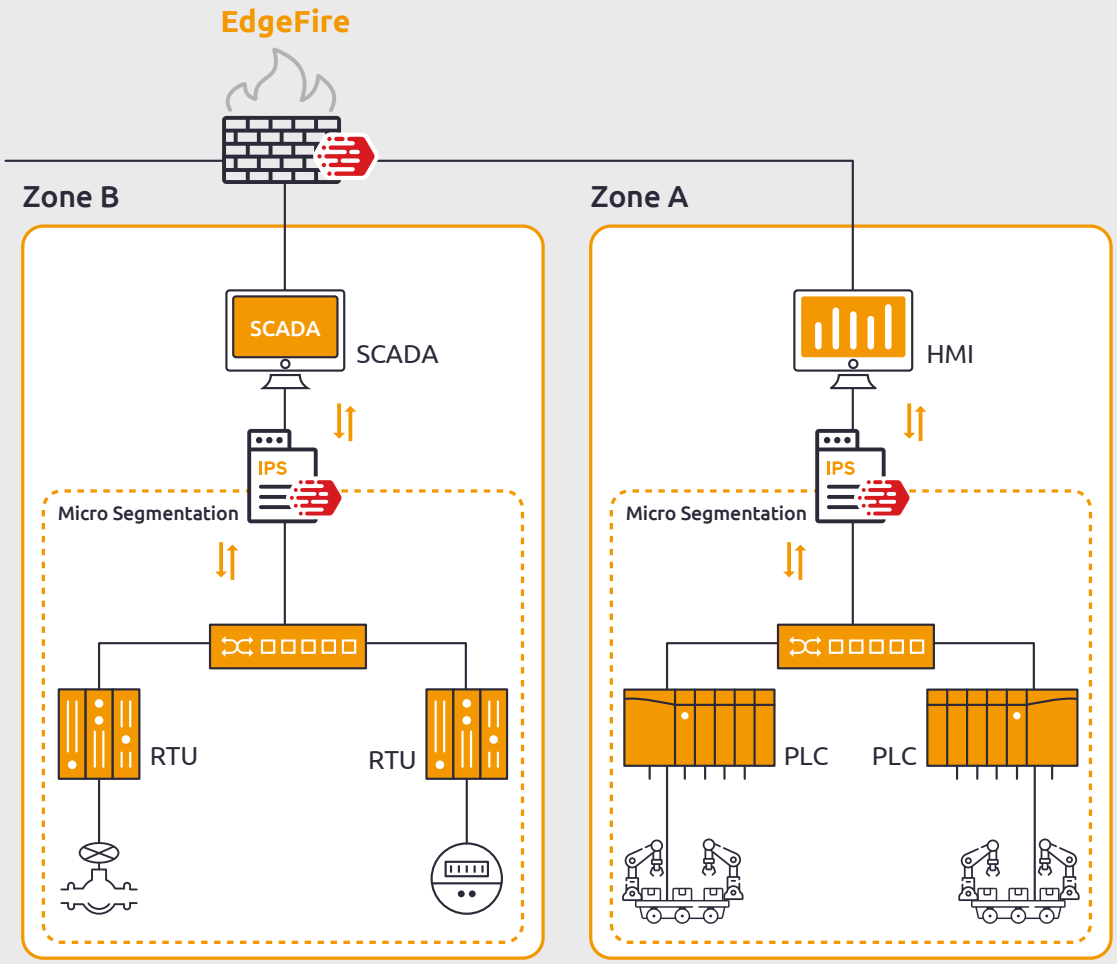
The link between IT and OT networks must be defended with OT firewalls designed specifically to decipher the protocols used by operational technology. Treacherous malware that can tunnel into your operations and shut down your production lines. Then, the hackers demand that you pay for encryption keys to unlock your systems. 15% of ransomware victims never recover their data, and hackers sometimes also succeed in encrypting backup data.

OT zero trust firewalls are smart. They know what's going on and will only allow trustworthy messages based on your specific manufacturing equipment and your security policies. Placing one OT firewall between your operational technology and IT is a good start. You may also want to create secure work cells by using OT firewalls to segment your network. Segmenting your network follows the theory of ship building – a ship with a single hull can be sunk by one leak, but a submarine with compartments in the hull can stay afloat. Each compartment can be sealed off, greatly mitigating threats to operations.

The OT Zero Trust Approach

TXOne's specialists recommend the following steps in order to protect your equipment and thereby your SMB:

1. **Deploy an OT next-generation firewall between your IT and OT systems.**
2. **Use a next-generation firewall or IPS to define network segments, safeguarding critical systems.**
3. **Deploy OT endpoint protection on all manufacturing equipment that connects to the networks.**
4. **Use OT zero trust handheld scanning devices to inspect each piece of new equipment and wipe away supply chain threats. Routinely scan legacy equipment to reinforce protection.**
5. **Gain moment-to-moment situational awareness by viewing your OT defense console.**





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